Dockerized AI Marketing Planning Service

Course Name: DevOps

Institution Name: Medicaps University – Datagami Skill Based Course

Student Name(s) & Enrolment Number(s):

Sr no	Student Name	Enrolment Number
1.	Kavish Jha	EN22CS301494
2.	Kritika Purohit	EN22CS301527
3.	Kunal Patidar	EN22CS301538
4.	Jatin Patidar	EN22CS301458
5.	Javesh Khargonkar	EN22CS301460
6.	Krishna Pawar	EN22CS301524

Group Name: Group 06D5

Project Number: DO-06

Industry Mentor Name: Mr. Vaibhav

University Mentor Name: Prof. Ritesh Joshi

Academic Year: 2025-26

Problem Statement & Objectives

1. Problem Statement

In modern organizations, marketing planning requires data-driven insights and automation. However, traditional systems lack scalability, automation, and easy deployment. Additionally, managing AI-based applications with database integration is complex without proper DevOps practices.

The challenge was to build and deploy an AI-powered marketing planning system that is scalable, portable, and accessible over the internet using modern DevOps tools and cloud infrastructure.

2. Project Objectives

- To develop an AI-powered marketing planning assistant using LLM (Gemini API)
- To containerize the application using Docker for portability
- To orchestrate services using Docker Compose
- To deploy the application on AWS EC2 for public access
- To integrate MySQL database for persistent storage
- To manage environment variables securely using .env
- To demonstrate DevOps practices such as deployment, debugging, and service management

3. Scope of the Project

- Development of AI-based marketing plan generator
- Backend integration with MySQL database
- Containerized deployment using Docker
- Cloud deployment using AWS EC2
- Basic Kubernetes setup (optional/experimental)
- Public accessibility via IP address

Proposed Solution

1. Key Features

- AI-based marketing plan generation using Gemini API
- Interactive UI using Streamlit
- MySQL database integration for storing plans
- Dockerized application for consistent deployment
- Multi-container setup using Docker Compose
- Cloud deployment on AWS EC2
- Environment-based configuration using .env
- Public access via browser

2. Overall Architecture / Workflow

1. User accesses the application via browser using public IP
2. Streamlit frontend takes user input (marketing goal)
3. Backend sends request to Gemini AI API
4. AI generates a marketing plan
5. Plan is stored in MySQL database
6. Previous plans are fetched and displayed
7. Docker manages containers (app + database)
8. AWS EC2 hosts the entire system

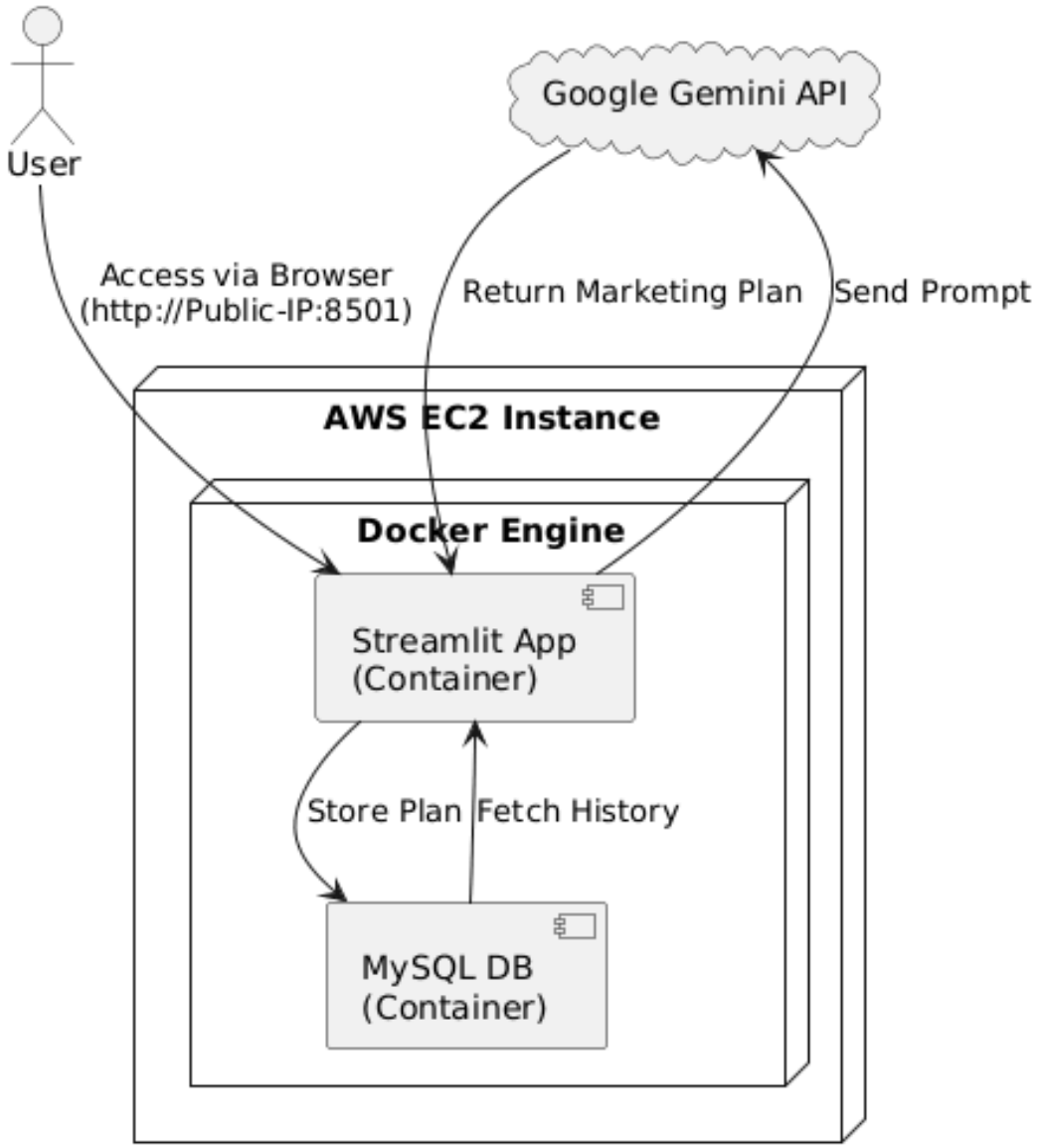


Figure 1: System Architecture of Dockerized AI Marketing Planning Service

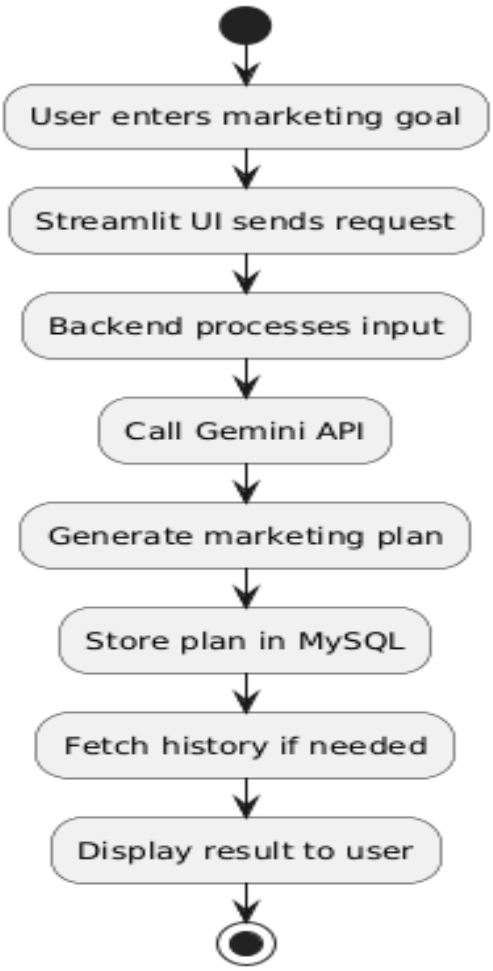


Figure 2: Workflow of AI-Based Marketing Plan Generation

3. Tools & Technologies Used

DevOps Tools

- Docker (Containerization)
- Docker Compose (Multi-container orchestration)
- Kubernetes (Basic deployment understanding)
- Git & GitHub (Version control)
- AWS EC2 (Cloud deployment)

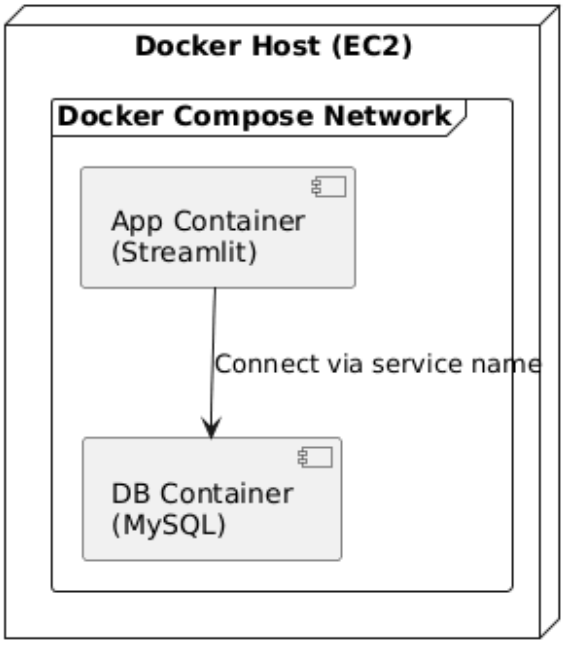


Figure 3: Docker-Based Deployment Architecture

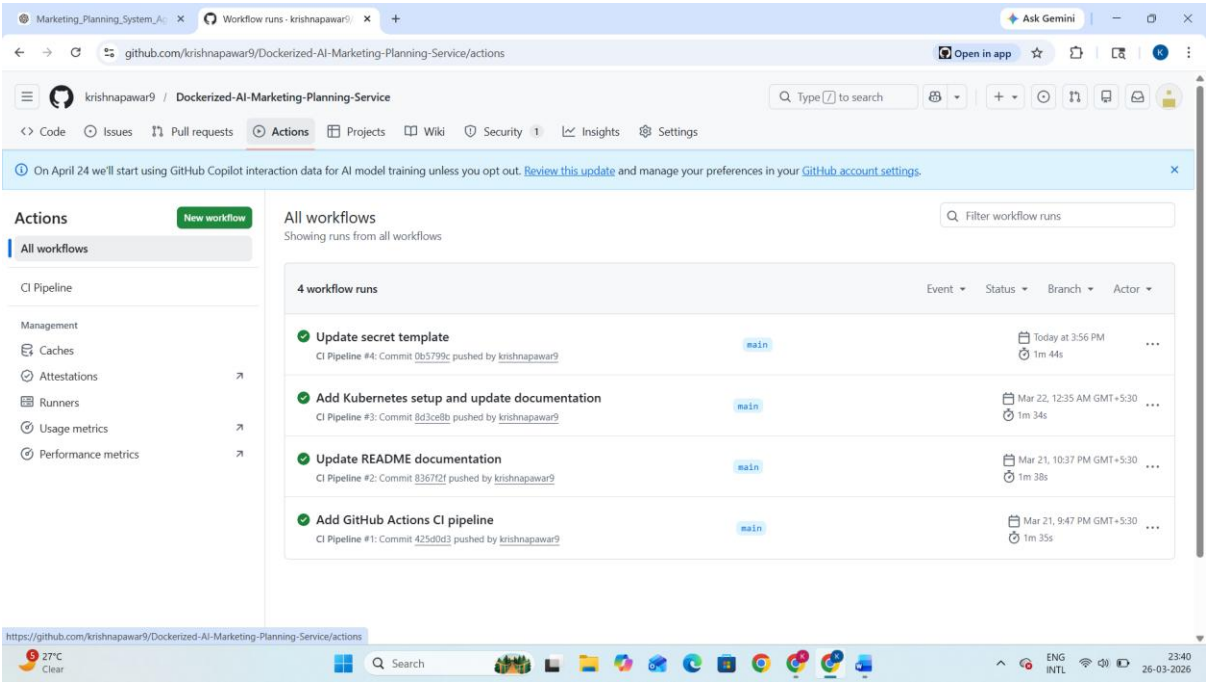


Figure 4: GitHub Actions CI/CD Pipeline Execution Dashboard

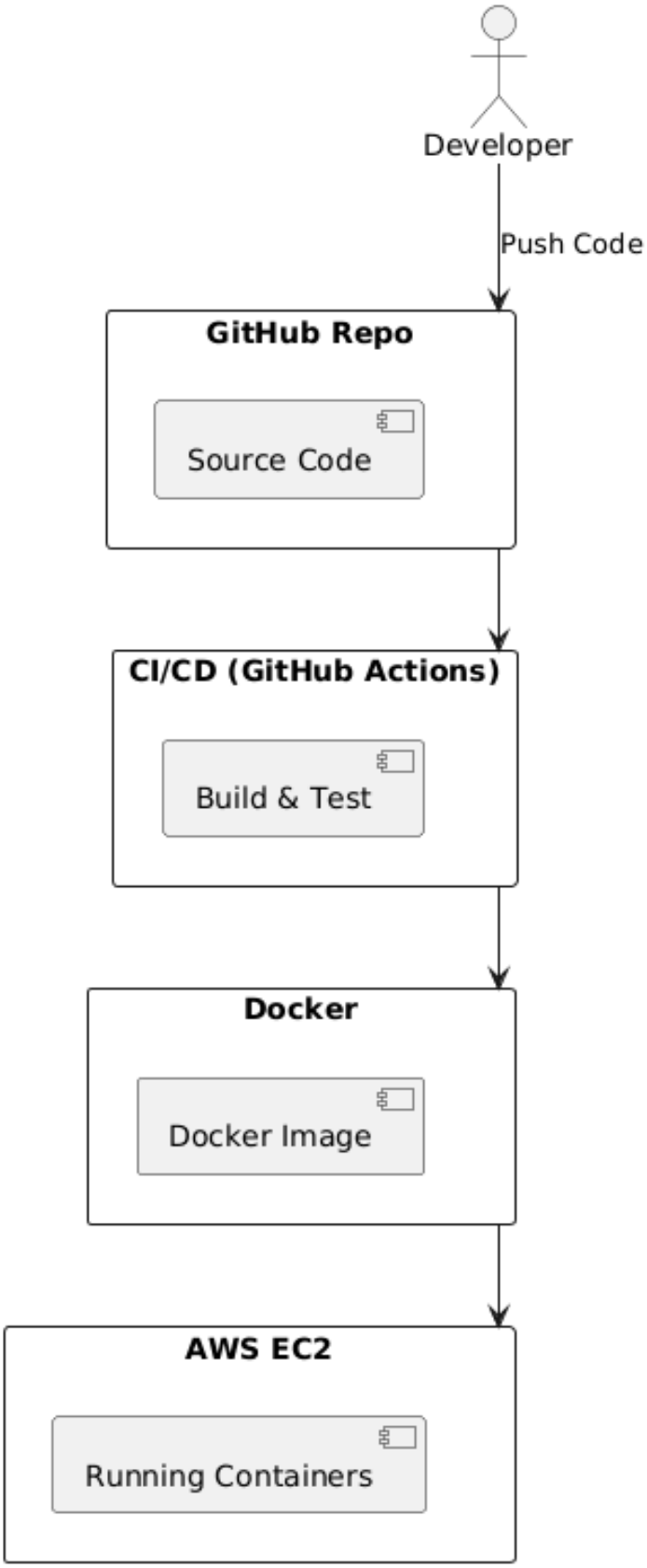


Figure 5: DevOps Workflow and CI/CD Pipeline

Backend & AI

- Python
- LangChain
- Google Gemini API

Frontend

- Streamlit

Database

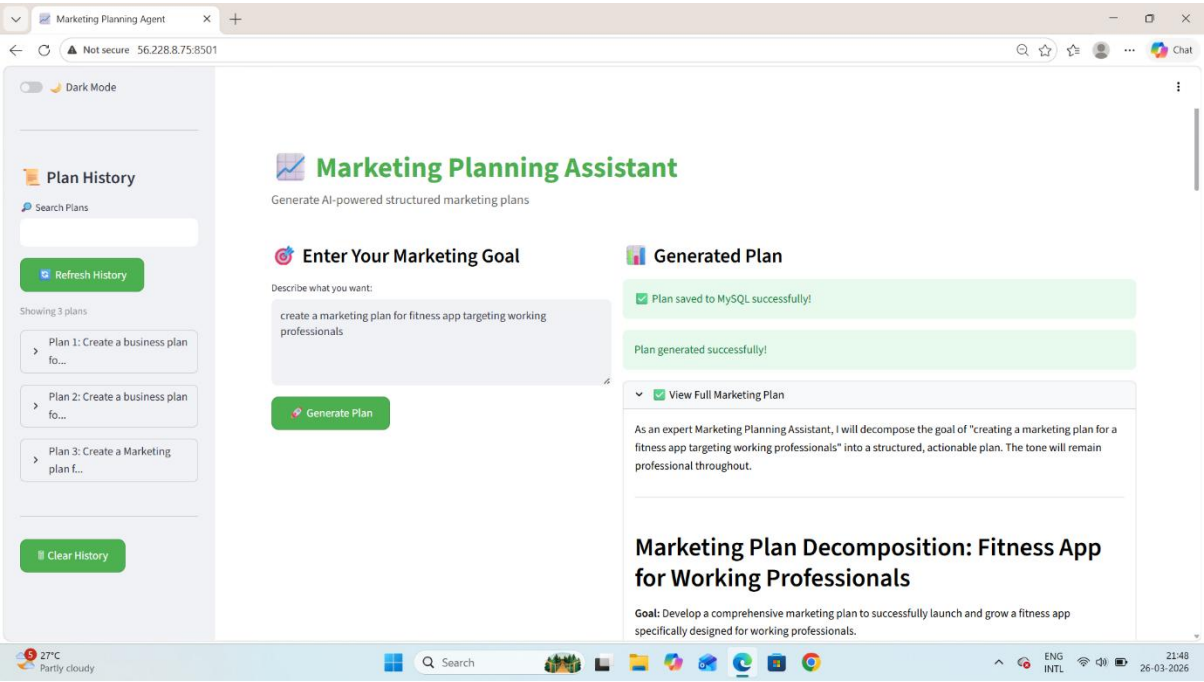
- MySQL

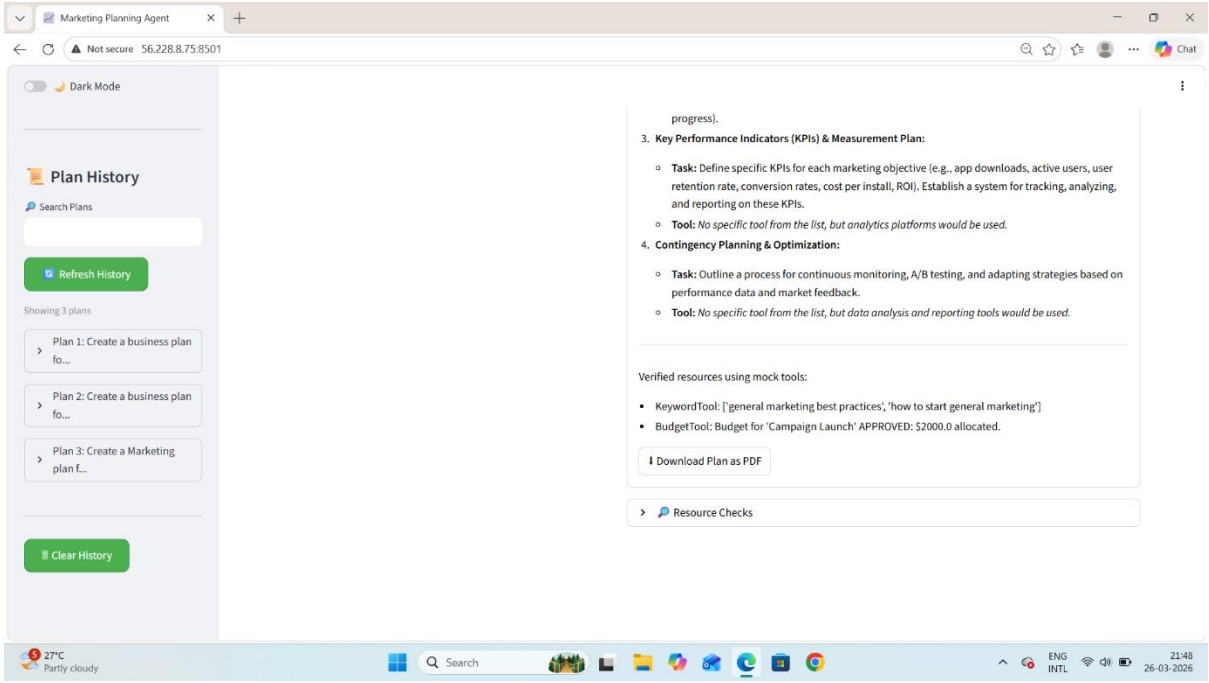
Other

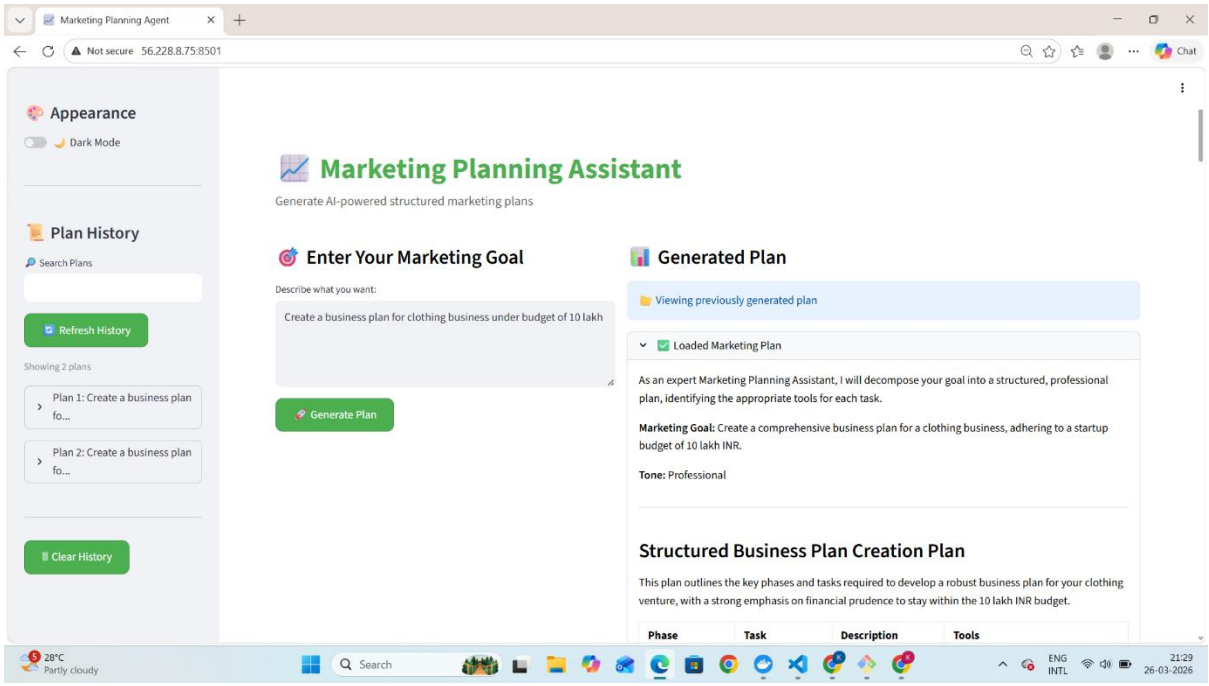
- Linux (Ubuntu EC2 instance)
- SSH (Remote access)
- Environment Variables (.env)

Results & Output

1. Screenshots / Outputs







- Running application on browser (AWS public IP)

2. Key Outcomes

- Successfully deployed a containerized AI application on AWS
- Achieved public accessibility via cloud deployment
- Implemented DevOps practices for real-world deployment
- Ensured portability using Docker
- Demonstrated service orchestration using Docker Compose
- Resolved real-time issues such as:
  - Container networking
  - API key management
  - Database connectivity

Conclusion

This project successfully demonstrates the integration of AI and DevOps practices. The application was developed, containerized, and deployed on AWS EC2, making it accessible over the internet. The use of Docker and Docker Compose ensured consistent environments and simplified deployment. The project also provided hands-on experience in handling real-world DevOps challenges such as environment management, service communication, and cloud deployment.

Overall, the project highlights the importance of DevOps in building scalable, reliable, and production-ready applications.

Future Scope & Enhancements

- Integration of multiple AI models (e.g., OpenAI, Claude) for better comparison and accuracy .
- Addition of user authentication and role-based access control.
- Advanced analytics dashboard for campaign performance tracking.
- Automated rollback strategy in case of deployment failures.